Influence of Cognitive Load on Voice Production: A Scoping Review

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Summary: Cognitive-motor interactions in speech production have a strong theoretical basis. However, majority of the existing literature has primarily focused on subjective and objective measures related to speech and not voice. This systematic review gathered evidence on the potential relationship between cognitive load and voice production. A search of five databases, website, citation review, and author search were completed in a sequential order to find published and unpublished literature from 1992 to 2022 using a combination of search terms including *voice, cognitive load/demand/effort/flexibility, dual task, and speech production/motor*. Studies for which the primary dependent variables were linguistic, or speech measures were included if voice acoustics was also measured and described. A final sample of nine articles were identified as meeting inclusion criteria: completed between 1992 and 2022, healthy adults (18+), and American English speakers. The review indicated that existing literature on the influence of cognitive load on voice production is limited. Acoustic measures, such as fundamental frequency, sound pressure level, and cepstral peak prominence, do not show consistent patterns of change with an increase in cognitive load. It is likely that the inconsistencies in the speech or cognitive task type and measurement of individual reaction to cognitive load changes may have led to these varied results. Further research using a range/continuum of cognitive tasks varying in load/difficulty level and physiological measurements is warranted to understand the underlying mechanisms of behavioral performance with implications for clinical voice assessment and rehabilitation.

Key Words: Voice production–Cognitive load–Dual tasks–Acoustics.

INTRODUCTION

Voice production is a complex interaction between the respiratory, phonatory, and resonatory systems. It is important that coordination exists between each of these subsystems for healthy voicing to occur. Changes in voice production can be the result of structural, neurological, and physiological changes associated with aging or with various pathologies such as laryngeal cancer and Parkinson's disease affecting any one or combination of these systems.

Recently, there has been a growing interest in investigating how factors such as stress, emotion, and cognition contribute to voice changes. Prior research on the relationship between cognition and voice production examined cognitive stress,[1-3] cognitive fatigue,[4] and cognitive load level.[5-7] For example, speaking in front of a large group, aka public speaking has shown to increase cognitive stress, and continuous or long task completion has shown to increase both physical and cognitive fatigue. This review will focus on cognitive load level since a change in cognitive load may influence and increase both stress and fatigue. Cognitive load theory (CLT[8]) describes how working memory influences learning and task completion. Cognitive load may be categorized as high or low based on the utilization of mental resources, such as attention, memory, and reasoning, and measured by accuracy of target task performance while completing a secondary task (eg, dual-task paradigms); however, the amount of cognitive or mental effort required will be dependent on the person's background and the function of the information to be learned.[9]

One method of assessing the role of cognitive load on voice production has been acoustic measurement using conventional measures such as fundamental frequency (*f*o), intensity, perturbation, noise, and spectral tilt.[10-12,6,13,1,2] Results have been inconsistent with no systematic increases or decreases in each of these measures with respect to an increased cognitive load. Some of the key reasons for this discrepancy include small sample size, methodological differences in task type, complexity of the tasks, cognitive domains assessed, defined or undefined reasoning for task selection, and inconsistencies in defining what qualifies as high and low cognitive load.[14]

The purpose of this systematic search and scoping review is to identify and synthesize acoustic measures in the context of the methodologies used to evaluate the relationship between cognitive load change and voice production in healthy adults speaking American English. The study objectives were to (a) review tasks used to assess voice production changes secondary to changing levels of cognitive load, (b) review cognitive tasks varying in load levels used in literature, (c) investigate how voice acoustic measures change with increasing cognitive load levels, and (d) identify knowledge gaps to develop aims for future research to better understand the role of cognition and voice production and its implication for clinical assessment and treatment.

METHODS

Data sources and search strategies

An initial search of peer-reviewed journals was completed between February 7 and 13, 2022, in PubMed, Medline, Web of Science, and Science Direct, using synonyms of

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TABLE 1.
Initial Search Terms for Various Databases

PubMed	Medline (CSA)	Web of Science	Science Direct	CINAHL
Voice	Voice		Voice acoustics	Voice
Speech and voice acoustics	Speech and voice measures	Speech motor		
Speech production				
Cognitive load	Cognitive load	Cognitive load	Cognitive load	Cognitive load
Dual task				
Adults	Adults		Adults	Adults

TABLE 2.
Secondary Search Terms

PubMed	Google Scholar
Voice; Speech and voice acoustics; Speech production	Voice production
Cognitive load; Cognitive demand; Cognitive effort; Cognitive flexibility; Dual task	Cognitive load
Adults	

important terms to represent the research question including voice, cognitive load, and adults (Table 1). These databases were chosen for content in biomedical, life sciences, allied health, and interdisciplinary resources.

Additional searches were completed between February 25 and 27, 2022, and March 15 and 17, 2022, secondary to a concern that the broad initial search may not have captured all the relevant studies, due to the small number of articles accepted for study inclusion. Therefore, a citation review of MacPherson et al[15] and Abur et al[16] was completed because they included multiple voice acoustic outcomes, the same cognitive task across age groups, and included autonomic arousal as an indicator of cognitive load. A search in PubMed for authors Christopher Dromey, Cara E. Stepp, Jason A. Whitfield, Charles Larson, and Susan Kemper, who are known to have completed research in related areas with healthy and disordered populations, was also completed. Additionally, a search of google scholar was completed using the terms "voice production and cognitive load" with the first hundred articles populated screened for inclusion with procedures described in Study selection. A final search was completed between April 1 and 5, 2022, in PubMed with an expansion on the search term "cognitive load" with appropriate synonyms, to retrieve articles that may have been missed in the initial search (Table 2). Search terms for voice outcomes were not expanded to incorporate specific acoustic terms, such as fundamental frequency, secondary to limited initial results with "voice" and "voice acoustics." Furthermore, the initial search provided results that were specific to individual acoustic measures, and adding any more specific terms may have restricted the number of articles retrieved.

Study selection

Initial search terms were used in the database listed above, and each article's title and abstract were reviewed, following the removal of duplicates, to create a reduced list for full text review. Articles were included on this reduced list if they noted combinations of the following key concepts related to the research question: speech and cognitive tasks, acoustics measures, and adults. Articles were excluded if the title and abstract indicated voice perception rather than voice production and inclusion of neurological conditions in the sample population. If there was a question as to the article's inclusion, it was added to the reduced list for further review of the full test. The same procedures were completed for the additional database, citation review, and author searches. For google scholar, only the first 100 articles populated were screened in this manner.

During the full text review, articles were assessed for the following inclusion criteria: completed between 1992 and 2022, including healthy adults (18+), located in the United States of America, and American English speakers. Only studies completed within the past 30 years were included as a longer time period may cause results to be affected by generational differences in participants, Furthermore, the cognitive load levels may be different due to varied participant experiences and living environments across a longer time period. Geographical and language criteria were included due to the known acoustical differences across languages.[17-19] As the primary outcome of interest was voice acoustics, only one language was included to examine fine-grained systematic differences in that language. Dependent variables of interest were voice acoustic measures and independent variables of interest were cognitive load activities. Studies whose primary dependent variables were linguistic, or speech, were included if voice acoustics were also measured and described. Studies were excluded if it was of an unrelated topic or did not include all topics of interest including speech and cognitive tasks and voice acoustic measures in healthy adults. Articles were excluded if they examined participants with neurodegenerative disease, cognitive impairment, laryngeal pathology, or if the study was completed in another language other than English and participants were not American English speakers.

Quality analysis

Concepts described by both Thompson et al[20] and Moeller et al[21] were used as a basis to complete an informal quality review on all articles retrieved for full text review. Both studies focus on reviewing articles for study design. Moeller et al[21] describes the application of the Horner quality indicator criteria to single study case design. The Horner Criteria[22] includes seven quality indicators: description of participants and setting, dependent variable, independent variable, baseline condition, experimental control/internal validity, external validity, and social validity, with these seven indicators having 21 subcomponents. Thompson et al[20] describe quality indicators to evaluate correlational studies. These include examining measurement, practical and clinical significance, avoidance of some common analytical mistakes, and confidence intervals for score reliability coefficient statistics and effect sizes. From these collective measures, full text articles were reviewed for reliability and validity of what and how variables were measured as compared to stated aims (measuring outcomes of interest), description of operational definitions of all dependent and independent variables, specifications outlined for how cognitive load was altered between tasks and how voice production was measured, and statistical information reported and interpreted for all dependent variables. Information was charted within an excel document for an overview of each study. No studies were excluded based on quality criteria since this scoping review aimed to present an overview of the existing literature.

Coding

Data tracking was maintained in an excel spreadsheet documenting citation, keywords, publication type, research question or aim, hypothesis, research design, number of participants, location, age, gender, cognitive status, groups, methods, cognitive and speech stimuli, control, analysis, results, significant effects, conclusions, and implications. Additional charting of cognitive tasks was completed in a separate excel spreadsheet for activity procedures, cognitive domain(s) being taxed, speech task, and modality of tasks including visual, verbal, auditory, and motoric. Decisions for these classifications were determined by the research team based on the task procedures described within each article.

RESULTS

Search results

A flowchart delineating the literature search is shown in Figure 1 following the 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Guidelines.[23] The database search yielded 420 records after removing duplicates, with 398 removed following title and abstract review, leaving 22 records sought for full text

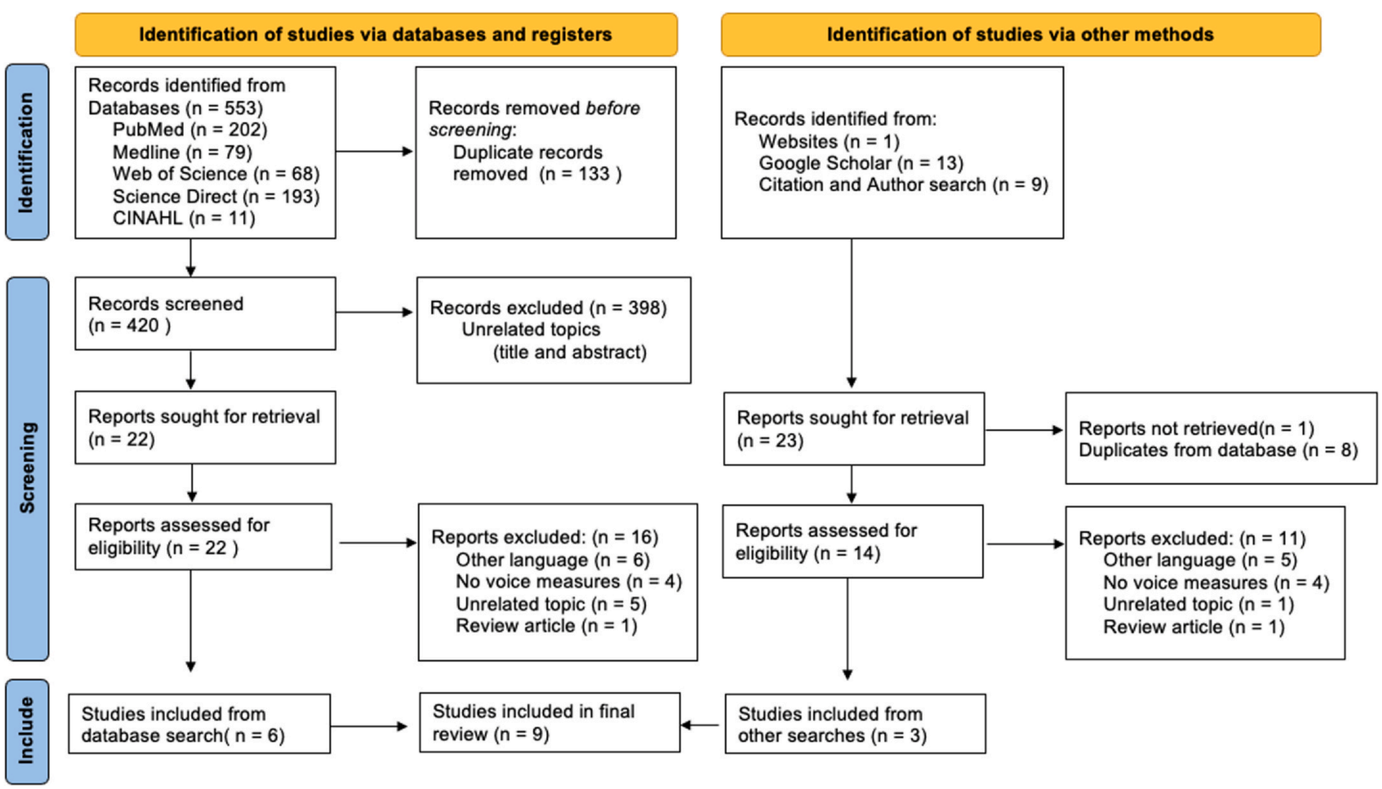

FIGURE 1. Flowchart depicting the identification of studies.

TABLE 3.
Participants' Characteristics

Study	Sample Size	Male/Female (%Male)	Average Age (Range)
Abur et al[16]	N = 12	6/6 (50%)	73 (68–78)
Cohen et al[26]	N = 281	134/147 (48%)	19.9 (18–64)
Dahl and Stepp[27]	N = 20	10/10 (50%)	20 (18–22)
Dromey and Bates[10]	N = 20	10/10 (50%)	Males 24 Females 22
Dromey and Shim[11]	N = 20	10/10 (50%)	Males 28 Females 21
Heaton et al[4]	N = 29	24/5 (83%)	20 (18–55)
Lee and Redford[24]	N = 20	13/7 (65%)	"College aged"
Lively et al[25]	N = 5	5/0 (100%)	"Graduate student" and "laboratory staff"
MacPherson et al[15]	N = 16	8/8 (50%)	25 (22–32)

review. The additional two searches resulted in 23 records but only 22 were able to be retrieved for full text review, with eight removed as duplicate records from original search. A total of 36 records were examined for study eligibility per inclusion criteria and 27 records were excluded, leaving nine records to be included in the final analysis. Of the excluded records, 11 were excluded for languages other than American English, eight records did not include voice acoustic outcome measures, six records were of an unrelated topic, and two were review articles.

Participants

Sample size ranged from 5 to 281 (Mean = 20, Mode = 20) with 78% of studies falling at or below 20 participants (Table 2). Mean age across studies was 22 years but spanned 18–78 years. Two studies did not include specific age demographics but described participants as "college age"[24] or "graduate students" and "lab staff,"[25] These descriptions were judged to fall within the stated age criteria for inclusion. Six studies were equally divided between male and female participants with a greater percentage of males in the remaining three studies Table 3.

Participant screening

Control for participant's baseline speech, language, hearing, cognitive, vision, or medical history was inconsistent. Below is a breakdown of the screening procedures completed. One study, Cohen et al,[26] did not report any initial participant screenings.

Speech and language

Four of the nine studies reported only participant self-report of no history of speech or language disorders.[10,11,24,27] Specifically, Lee & Redford[24] reported no speech or reading disorders, and Dahl & Stepp[27] reported no communication disorders, which was interpreted to include both speech and language. Two studies reported only participant self-report of no speech disorders[4,25], with Heaton et al[4] also reporting self-report of no facial muscle disorder that may impact speech production. MacPherson et al[15] also described participant self-report of no speech or language disorders; however, this study along with Abur et al[16] also used objective measures in these areas. Both studies completed the Oral Mechanism Screening Examination–3rd Edition, the word identification and passage comprehension subtests of the Woodcock Reading Mastery Tests–Revised, and the sentence reading subtest of the Psycholinguistic Assessment of Language Processing in Aphasia.

Hearing

Two studies did not report screening hearing status.[4,26] Two of the nine studies only relied on participant self-report of no hearing disorder.[24,25] Of the remaining five studies, two reported hearing status through pure tone hearing screening only[16,27] and three used both participant self-report followed by the completion of a pure-tone hearing screening.[10,11,15]

Cognition

Screening for participant cognitive status was also inconsistent with only four studies reporting screening in this area. Two studies only used participant self-report of no cognitive disorders[27] and no neurological disorders, which is included here as this may influence cognitive status.[4] Both Abur et al[16] and MacPherson et al[15] used the Cognitive Linguistic Quick Test to objectively measure cognitive status. Five studies did not report screening for cognitive status.[10,11,24–26]

Vision

Three of the nine studies screened for color blindness using the Ishihara Color Blindness Test.[16,27,15] The remaining six studies did not report any vision screenings.

Sleep disorders

Three of the nine studies used client self-report of sleep disturbances including sleep apnea. Abur et al[16] did not provide a time frame for the self-report; however, MacPherson et al.[15] stated that there was no reported sleep disturbance or sleep apnea in the 6 months preceding participant's participation in the study. Heaton et al[4] also did not provide a time frame for the self-report but utilized the Epworth Sleepiness Scale to assess daily fatigue levels of participants and asked participants to get 7 hours of sleep per night during participation in the study. The remaining six studies did not report screening for sleep disorders.

Mood and/or psychological disorders

Only two of the nine studies reported screening for mood or psychological disorders. Both Abur et al[16] and MacPherson et al[15] used client self-report of no depression, anxiety, or other psychological disorders. MacPherson et al[15] further specified that these disorders did not occur within 6 months prior to participation in the study. The remaining seven studies did not report screening for mood or psychological disorders.

Tobacco use

Three studies screened for a history of tobacco use. Participant self-report was utilized by Heaton et al[4] for a history of tobacco product use, while Abur et al[16] and MacPherson et al[15] screened for no smoking within 5 years prior to participation in the study. No screening for tobacco use was found in the remaining six studies.

Medical history

Three studies included a medical history participant self-report. Heaton et al[4] questioned cold, cough, or sinus disorders at time of participation and any history of neurological disorders. Additionally, Heaton et al[4] along with Abur et al[16] and MacPherson et al[15] asked participants about disorders associated with the upper extremities. Heaton et al[4] specified injury to the hand and finger range of motion, while Abur et al[16] and MacPherson et al[15] reported no sensorimotor disorders of the upper extremities. Abur et al[16] and MacPherson et al[15] also both used client self-report of no dermatological conditions or high or low blood pressure. MacPherson et al[15] completed additional medical review through client self-report that included screening for conditions known to effect autonomic failure such as a history of autonomic failure, multi system atrophy, diabetes, chronic obstructive pulmonary disease, emphysema, history of drug and alcohol abuse, and schizophrenia. Specific to the 6 months prior to participation in the study, MacPherson et al[15] also asked questions regarding pregnancy and nursing, and prediabetes or other metabolic syndromes.

Medications

MacPherson et al[15] was the only study to report client self-report medication use. It was reported that clients were asked questions about medication use specific to those that effect motor and cognitive function, including medications for attention deficit disorder, anticonvulsants, and muscle relaxers.

Requirements prior to data collection

Both Abur et al[16] and MacPherson et al[15] asked that participants refrain from alcohol, caffeine, large meals, and heavy physical activity 3 hours prior to participation. Heaton et al[4] asked that participants agree to get 7 hours of speech per night during study participation.

Cognitive and speech tasks

Heterogeneity was seen in speech and cognitive tasks used to manipulate cognitive load (Table 4). Six studies contained dual-task activities including read sentences, free speech, or word-level productions in combination with a secondary linguistic, memory, or visual-motor task. There was no indication of method for choosing dual-task pairs or which cognitive domain was being manipulated. In multiple studies, more than one task was completed, but acoustic outcomes were not reported based on task. The remaining three studies used a single sentence-level Stroop task in a congruent (low level) or incongruent (high level) condition. Abur et al[16] and MacPherson et al[15] followed the same procedures while Dahl and Stepp[27] adjusted sentence structures to measure different acoustic outcomes.

Individual cognitive load changes and autonomic arousal

Two studies included objective measures of autonomic arousal to examine its relation to voice production in congruent and incongruent conditions of the sentence-level Stroop task.[15,16] In younger adults (Mean age = 25 years), MacPherson et al[15] found autonomic arousal was predictive of cognitive load, with pulse volume amplitude lower and pulse period shorter during incongruent conditions. Abur et al[16] completed the same task with older adults (Mean age = 73 years) and found that the two autonomic arousal measures were predictive of load condition. Skin conduction response amplitude was higher and pulse volume amplitude was lower in incongruent conditions. Contrary to the younger age group, there was no difference in pulse period between the two conditions in the older age group.

Voice outcomes

Voice outcomes were highly variable due to differences in speech tasks (eg, phonation, single digit or word, sentence repetition, or spontaneous speech). The most common voice outcomes reported were fundamental frequency (fo)—an acoustic correlate of vocal pitch, sound pressure level (SPL)—an acoustic correlate of loudness, and cepstral peak prominence (CPP)—a measure of overall severity of dysphonia. Results are summarized in Table 4.

TABLE 4.
Cognitive and Speech Tasks

Author (Year)	Tasks	Domain	Task Presentation	Output Modality
Abur et al[16]	Sentence-level Stroop task	Selective attention; executive function/ inhibition; language	Visual	Verbal
Cohen et al[26]	*Dual task:* Free speech and nonlinguistic memory recall	Selective attention; working memory; language; executive function/inhibition	Visual	Verbal, motoric
Dahl and Stepp[27]	Sentence-level Stroop task	Selective attention; executive function/ inhibition; language	Visual	Verbal
Dromey and Bates[10]	*Isolated tasks:* speech only, linguistic only, cognitive only, visuomotor only	Dual task: language; executive function; visuospatial; attention; working memory	Verbal	Verbal
	Dual task: speech only task combined with each other task		Visual	Motoric
Dromey and Shim[11]	*Isolated task:* speech sentence repetition task and one word association task	Dual task: working memory; attention; executive function; language	Verbal	Verbal
	Dual task: speech repetition and manual motor skill task		Visual	Motoric
Heaton et al[4]	*Dual task:* nonlinguistic visual motor task with reading/ repeating sentences, paced auditory serial addition test; and digit span recall	Dual task: visuospatial; motor; language; executive function; recall	Visual	Verbal; motoric
Lee and Redford[24]	*Alternating tasks:* Reading sentences and linguistic memory task	Working memory; attention; language	Visual	Verbal
Lively et al[25]	*Dual task:* repeated phrases with nonlinguistic visual tracking task	Dual task; visuospatial; attention; language	Visual	Verbal; motoric
MacPherson et al[15]	Sentence-level Stroop task	Selective attention; executive function/ inhibition; language	Visual	Verbal

Fundamental frequency

Six studies analyzed fundamental frequency as a primary voice outcome. Three of these used similar procedures for sentence-level Stroop tasks with both congruent and incongruent conditions,[16,27,15] but no significant relationship between cognitive load and *f*o was reported. Results were inconsistent in the dual-task paradigms with respect to the selection of specific secondary tasks, where participants were asked to complete speech tasks with a co-occurring linguistic, motor, or visuospatial task. Lee and Redford[24] found no relationship between cognitive load condition and *f*o during reading sentence and co-occurring *linguistic* memory tasks (complex letter or spatial location recall from a choice of eight options) but reported a significant effect of relative clause type and location in read sentences on *f*o variability. In contrast, Cohen et al[26] used free speech and a *non-linguistic* continuous performance visual recall task of pressing the space bar for a target stimulus while inhibiting distractors, and found decreases in frequency, perturbation, and *f*o variability. Lively et al[25] used a repeated phrase and visuospatial task of keeping a pointer centered between two boundaries. However, due to inconsistent changes in *f*o among the five participants, they were unable to compare *f*o results or make conclusions about the nature of its change.

Sound pressure level

SPL was an outcome measure for dual tasks in six studies that contained both verbal and limb motoric output during tasks. Cohen et al[26] noted a decrease in SPL associated with cognitive demand during free speech and continuous performance visual memory and inhibition tasks, but Dromey and Bates[10] and Dromey and Shim[11] found increases in SPL in concurrent speech and motor tasks meant to divide participants attention in both group and individual comparisons ($P < 0.001$). Both studies used both single- and dual-task paradigms. Dromey and Bates[10] had participants complete four isolated tasks including sentence repetition, generation of grammatically correct sentences given eight to nine words, two-digit math subtraction, and a visuomotor tasks of clicking on a moving target. Sentence repetition was then paired with each of the other three tasks. Dromey and Shim[11] had participants complete repeated sentences, the One Word Association Test for verbal fluency, and a pegboard motor task with both dominant and non-dominant hand. They then paired one trial with each hand on the pegboard task with either of the speech tasks. Neither Abur et al[16] nor MacPherson et al[15] found a relation between condition and SPL in older and younger populations using the sentence-level Stroop task in either the congruent or incongruent conditions. As with *f*o, Lively et al[25] were unable to compare SPL data due to the inconsistencies noted among their five participants while repeating phrases and keeping a target centered between boundaries.

Cepstral peak prominence

Abur et al[16] and MacPherson et al[15] assessed CPP using the same procedures for a sentence-level Stroop task in both congruent and incongruent conditions. MacPherson et al[15] found that increased CPP was significantly associated with higher load conditions, but Abur et al[16] did not find this in an older population. For this group, CPP remained consistent across conditions.

Other acoustic measures/outcomes

Three other voice outcomes were measured within the reviewed studies. Dahl and Stepp[27] found a large effect (0.79) during sentence-level Stroop tasks for cycle on mean relative fundamental frequency (RFF) onset indicating a possible increase in laryngeal tension influencing vocal output when cognitive load increases. RFF measures short-term changes in *f*o in voice cycles immediately before (offset) and after (onset) a voiceless consonant normalized by the typical *f*o of the voiced phoneme before and after the consonant to estimate baseline laryngeal tension and reflecting vocal effort.[28] L/H ratio was found to be lower in higher load conditions also in sentence-level Stroop tasks by MacPherson et al.[15] For dual tasks, Heaton et al[4] examined vocal quality and found an increase in low and irregular pitch use which they reported as "vocal creak" probability using an algorithm based on *f*o while participants read sentences paired with serial addition and a visuomotor task of moving an object without touching the edges of the computer screen, and an isolated digit span recall task. Use of this aperiodic voice was found to be a significant predictor of cognitive change during numerical recall, and the paired reading and nonlinguistic motor tasks Table 5.

DISCUSSION

This scoping review was aimed to summarize the current state of knowledge and identify knowledge gaps in the current literature, regarding the influence of cognitive load change on voice production in healthy American English-speaking adults. Our systematic search led to the identification of nine studies that were reviewed for cognitive tasks, speech tasks, and acoustic measures.[14] From the results obtained, implications for further research are suggested—considerations on the development of a hierarchy and continuum of cognitive tasks varying in load levels, and how they might be beneficial for assessment, treatment, and early identification of voice and speech-motor control disorders such as those in people with neurodegenerative disorders (eg, Parkinson's disease).

Cognitive load and speech task development

Reviewed studies varied in terms of the cognitive tasks, task loads, and speech tasks selected, and this limited the comparisons across them, as well as contributed to the discrepancies in the results. Although speech and cognitive

TABLE 5.
Voice Acoustic Outcomes

Author (Year)	Sample Size	Tasks	Results
Fundamental frequency			
Abur et al[16]	N = 12	Sentence-level Stroop task	*Not significant* predictor of condition
Cohen et al[26]	N = 281	*Dual task:* Free speech and nonlinguistic memory recall	*Decreased* f o and *fo* variability
Dahl and Stepp[27]	N = 20	Sentence-level Stroop task	*No significant* difference in *fo* between conditions Small effect of condition on mean RFF offset (0.01) Medium effect of cycle on mean RFF offset (0.10) Large effect of cycle on mean RFF onset (0.79)
Lee and Redford[24]	N = 20	*Alternating tasks:* Reading sentences and linguistic memory task	*No effect* of condition on *fo* Significant effect of relative clause type on normalized measure of *fo* variation and pitch range across sentence Significant effect of relative clause location with respect to matric clause on *fo* variation and range Increased *fo* range and variability in sentences with relative clauses at the end of the matric clause
Lively et al[25]	N = 5	*Dual task:* repeated phrases with nonlinguistic visual tracking task	Inconsistent changes between participants; unable to compare across
MacPherson et al[15]	N = 16	Sentence-level Stroop task	*No difference* between conditions
Sound pressure level			
Abur et al[16]	N = 12	Sentence-level Stroop task	*Not significant* predictors of condition
Cohen et al[26]	N = 281	*Dual task:* Free speech and nonlinguistic memory recall	*Decreased*
Dromey and Bates[10]	N = 20	*Isolated tasks:* speech only, linguistic only, cognitive only, visuomotor only *Dual task:* speech only task combined with each other task	*Increase* in all dual-task conditions greater for speech + visuomotor combined task
Dromey and Shim[11]	N = 20	*Isolated task:* speech sentence repetition task and one word association task *Dual task:* speech repetition and manual motor skill task	*Increased* in concurrent motor task
Lively et al[25]	N = 5	*Dual task:* repeated phrases with nonlinguistic visual tracking task	*Inconsistent* changes between participants; unable to compare across
MacPherson et al[15]	N = 16	Sentence-level Stroop task	*No difference* between conditions
Cepstral peak prominence			
Abur et al[16]	N = 12	Sentence-level Stroop task	*Not significant* predictors of condition
MacPherson et al[15]	N = 16	Sentence-level Stroop task	*Significantly* associated with condition
Other acoustic			
Dahl and Stepp[27]	N = 20	Sentence-level Stroop task	Small effect of condition on mean RFF offset (0.01) Medium effect of cycle on mean RFF offset (0.10) Large effect of cycle on mean RFF onset (0.79)
Heaton et al[4]	N = 29	*Dual task:* nonlinguistic visual motor task with reading/repeating sentences, paced auditory serial addition test; and digit span recall	"Vocal creak" based on *fo* *predicted* cognitive performance decrease

tasks did not have stated reasons why or how tasks were chosen, working memory and attention seem to be the two primary cognitive domains considered within CLT that were consistently challenged. The input and output modalities within the dual tasks were also inconsistent, decreasing the ability to validate and compare results across studies. A clear description on selection of tasks, sensory-motor modalities, and how they influence cognitive load is crucial as some modalities may automatically be at a higher load (eg, visual vs motor). Most studies used laboratory-based tasks that were easier to complete and assess under the limitations of a research study. The use or simulation of tasks from everyday life would increase ecological validity and could help further dual-task development that would be more generalizable to the natural setting. Systematic investigation of this topic through comparison of tasks with similar design and modality should focus not only on input and output classification, but also on what cognitive domain(s) may have the most influence over voice production. Controls for cognitive status and other confounding variables that may influence cognitive performance should also be taken into consideration, as these were limited to participant report or undocumented in most studies. This understanding may assist in the creation of appropriate methodology, standardization, and reliability in our research task development that may then be generalized into the clinical setting.

Individual cognitive load

Without clear measures to determine individual participant changes in cognitive or mental effort, results cannot be generalized to groups or larger populations. Each participant may have a different background in education and life experience, and they may process and attend to activities in different ways. Therefore, a high cognitive load for one is not necessarily the same for another, influencing how participants may learn, complete, and produce voice during these tasks. This "not one size fits all" concept will need to be further systematically studied using a combination of physiological and neurophysiological measures so that the individual mechanisms underlying the behavioral performance can be thoroughly understood. Such a precision-based adaptive approach that can be customized to individual patients is essential in the field of disordered populations.

Autonomic arousal has been used as a physiological measure of cognitive load,[29] and two of the reviewed studies examined the relationship with voice production and found autonomic arousal to be predictive of cognitive load level.[15,16] Including measures of autonomic arousal and/or subjective participant surveys on the effort to complete tasks, as described in Esmaeili Bijarsari[30] and Dahl et al.[27], may validate these measures and allow for subjective reporting to be used clinically, when objective measures are not available. Further measurement of participant stress, emotion, and fatigue levels may provide additional information on how a combination of these psychological factors influence voice production, to understand under what conditions voice production is most likely changed.

Voice production outcomes

Voice production outcomes were limited to three main acoustic measures: *f*o, SPL, and CPP. Additional vocal quality measures included RFF and "vocal creak" which used *f*o in an algorithm to describe low and irregular pitch use; however, these were limited to one study each. Voice outcomes were also highly variable due to task differences such as reading vs sentence/word production. Although these tasks offer controlled linguistic and phonetic environments, they lack ecological validity because they do not reflect how a person uses their voice in functional conversational exchanges. Manipulating cognitive load in single tasks (eg, sentence-level Stroop tasks) did not affect *f*o or SPL, but CPP was significant in younger adults. Therefore, studies that examine measures of vocal quality may be more beneficial.

LIMITATIONS

The current review has limitations that must be taken into consideration. Multiple searches were completed secondary to the small number of articles initially retrieved and to be included in the final review. There was a concern that all relevant research may not have been captured, and therefore, variations on the original search terms were included in the subsequent searches. It could be argued that cognitive stress and fatigue may lead to increased cognitive load. However, this was not explicitly examined in this current study. Another limitation is that only one author independently completed the article selection for this scoping review.

SUMMARY

Existing literature on the influence of cognitive load on voice production is limited. Overall, no consistency was found in cognitive, or speech tasks and in the dual-task interference across studies in healthy adults who spoke American English. Only two studies examined objective measures of cognitive load change. Voice outcomes were highly variable secondary to task differences.

FUTURE DIRECTIONS AND CLINICAL IMPLICATIONS

Future studies on investigation of influence of cognitive load on voice production warrant (a) examination of task hierarchy (eg, simple to complex/difficult continuum using different or same tasks but on a fluid continuum/scale), (b) examination of multimodal assessment methodologies (eg, physiological indices of load such as autonomic arousal and neurophysiological indices of load such as functional imaging of cerebral blood flow), and (c) extension to disordered populations (eg, Parkinson's disease, multiple sclerosis, amylotrophic lateral sclerosis). Future studies also warrant diverse language contexts[31] to examine if the relationship between cognitive load and voice varies across languages. By understanding how changes in cognitive demand influence voice production, researchers can enhance clinical voice diagnostics, treatment, and carryover task development. Specifically, per clinical assessments, severity of deficits determined in a well-controlled clinical setting is not representative of a patient's everyday voice. This is in part because conventional clinical practice methods treat speech/voice as an isolated system, ignoring the interplay between speech/voice, motor, and cognitive systems. For example, routine clinical evaluations often require patients to say vowels, repeat sentences, and produce connected speech samples in isolation. This method leads to increased attention to the speech task, often resulting in improved performance and intelligibility. However, in real life, we often speak in the presence of other daily activities that tax our cognitive or motor systems. Therefore, clinical voice assessments that integrate/account for the interaction between cognitive and speech/voice systems may more accurately represent a patient's handicap in their everyday environment. Within treatment, understanding the interaction between impaired cognition and voice production may inform treatment design to best facilitate achievement of therapy goals. Development of a task hierarchy (cognitive, motor, speech/voice) will enable clinicians to either challenge or ease the cognitive burden systematically, potentially generalizing skills faster by increasing behavioral motor learning. Clinicians may also be able to better personalize treatment protocols based on cognitive domains, client reflections of the effort involved, and the outcomes being produced.

Declaration of Competing Interest

The authors declare no competing financial or nonfinancial interests related to this study.

No part of the submitted article is under consideration for publication in any other journal. Preliminary findings of this research were presented as a poster at the American Congress of Rehabilitation Medicine in November 2022, Chicago and an abstract was published in *Archives of Physical Medicine and Rehabilitation* in 2022.

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